

Shrivatsa Deshmukh

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Summary: Computational neuroscience and NeuroAI researcher with a background in robotics and physics, combining Bayesian inference, statistical modeling, and deep learning to study neural coding, circuit dynamics, and disease-related dysfunction from single neurons to whole-brain systems.

SKILLS

Neuro & Computational: EEG, fMRI, Calcium Imaging, Neural Data Analysis, Neural Mass Models, Simulation-Based Inference (Normalizing Flows), Hierarchical Statistical Modeling (GLMMs), MNE-Python

Programming & ML: Python, NumPy, SciPy, Pandas, PyTorch, Scikit-learn, MATLAB, Git, Linux/Bash

PUBLICATIONS

- **Shrivatsa Deshmukh**, Sasha Brenner, Nico Scherf, and Thomas R. Knösche. “Gate-Based Feature Importance in Simulation-Based Inference for EEG: A Cautionary Tale.” *NeurIPS 2026 Workshops- Interp4Discovery: Interpretability for Discovery and Sim2Science: ML with Imperfect Scientific Models* (accepted at both).
- **Shrivatsa Deshmukh**, Vitthal Khatik, and Anupam Saxena. “Robust Fusion Model for Handling EMG and Computer Vision Data in Prosthetic Hand Control.” *IEEE Sensors Letters* 2023. [doi:10.1109/LSENS.2023.3301837](https://doi.org/10.1109/LSENS.2023.3301837)

EXPERIENCE

Noel Lab, University of Minnesota | Research Associate

November 2025 – Present

- Designed a zero-inflated negative binomial GLMM (ZINB) pipeline to characterize ASD-associated neural signatures across 3 mouse models (*Fmr1*, *Cntnap2*, *Shank3* KOs) and wild-type controls in International Brain Laboratory (IBL) electrophysiological data spanning 32 brain regions.
- Quantified neural response variability, contrast response functions, and adaptation dynamics across 613 neurons and 375,000+ neuron-trials; preliminary ASD vs. wild-type differences align with existing literature, with cross-model analysis ongoing.

Brain Networks Group, Max Planck Institute CBS | Guest Researcher

May 2025 – September 2025

- Built a simulation-based inference pipeline (neural posterior estimation with a neural spline flow) to infer physiological parameters of a Jansen-Rit neural mass model from EEG simulated through a realistic head-model forward projection; $R^2 = 0.79$ with well-calibrated posteriors.
- Tested whether learned feature gates reveal which EEG features drive inference, benchmarking them against retrain-based ablation across seeds. Found that the gates, though stable across seeds and accuracy-neutral, were negatively correlated with ablation importance (Spearman $\rho \approx -0.6$) and down-weighted the most important feature, showing that learned weights are not a reliable readout of feature importance.

Alfonso Araque Lab, University of Minnesota | Research Associate

March 2025 – November 2025

- Analyzed AQuA2-segmented astrocyte calcium imaging (3,008 events, 48 recordings) to test how psilocybin alters astrocyte signaling in wild-type (WT), 5-HT_{2A}-antagonist (volinanserin), and IP₃R2 conditional-knockout slices.
- Found that psilocybin increased astrocyte integrated calcium signal by 25% and event duration by 20% (both $P < 0.001$); both effects were abolished in IP₃R2 knockouts ($P < 0.01$ vs. WT), while 5-HT_{2A} blockade trended toward attenuation but did not differ significantly from WT.

Health Intelligence Lab, University of Minnesota | Research Associate

November 2024 – March 2025

- Applied tensor decomposition to fMRI data from ~100 subjects to examine amygdala reactivity during an emotion-processing task; found unexpectedly weak amygdala-BDI (Beck Depression Inventory) correlations despite the amygdala’s established role in emotion processing.
- Explored dynamic mode decomposition (DMD) and 6 variants via PyDMD on resting-state EEG; identified Optimized DMD as most noise-robust and characterized mode incoherence across time windows as the fundamental limit of long-horizon decomposition.

Indian Institute of Technology Kanpur | Machine Learning Researcher

January 2023 – August 2023

- Designed an EMG-vision fusion model for prosthetic-hand grasp classification that uses object-detection output (YOLOv5) as a classifier input, making it robust to detection errors.
- Improved accuracy by up to 11.57% over prior fusion methods and, unlike them, stayed above EMG-only accuracy under 25% object-label corruption.

PROJECTS

Neural Phoneme Decoder for Speech BCI

January 2025 – May 2025

- Reproduced the gated recurrent unit (GRU) phoneme decoder from Willett et al. (2023, Nature); isolated the decoder stage from the full Kaldi pipeline and reduced compute requirements via hyperparameter optimization to enable single-GPU training, achieving 78.77% phoneme accuracy within 1.5% of the published benchmark.

EDUCATION

University of Minnesota – Twin Cities

Master of Science in Robotics (Specialization: Medical Robotics)

August 2024

GPA: 3.95 / 4.00

Birla Institute of Technology and Science (BITS), Pilani

Bachelor of Engineering in Mechanical Engineering

Master of Science in Physics

August 2018 – May 2023

GPA: 8.37 / 10.00